Job Acceptance Prediction System

Project Overview

The Job Acceptance Prediction System is a machine learning project designed to analyse candidate information and predict their job placement status based on historical data.

The project uses candidate attributes such as academic performance, technical skills, aptitude scores, communication skills, work experience, employment history and job-role suitability to identify patterns associated with placement outcomes.

By applying machine learning classification algorithms, the system learns from historical candidate records and predicts whether a candidate is likely to be placed or not placed.

The project also includes an interactive Streamlit dashboard that allows users to explore candidate information, understand model performance and generate predictions for new candidates.

The main objective is to demonstrate how machine learning can be applied to HR analytics to support data-driven placement analysis.

Objectives

The primary objectives of this project are:

1. Understand and analyse the HR job placement dataset.
2. Identify missing values, duplicate records and inconsistent data.
3. Perform data cleaning and preprocessing.
4. Conduct Exploratory Data Analysis (EDA) to understand candidate characteristics and placement patterns.
5. Perform feature engineering to create meaningful additional features.
6. Apply different machine learning classification algorithms.
7. Compare model performance using appropriate evaluation metrics.
8. Select and optimise the best-performing model.
9. Develop an interactive Streamlit dashboard.
10. Allow users to enter candidate information and generate placement predictions.

Dataset Description

The project uses the HR_Job_Placement_Dataset.csv dataset.

The dataset contains candidate-related information, including academic qualifications, assessment scores, employment history, job preferences and placement status.

The initial dataset contains:

Description	Value
Total records	51,500
Total columns	26
Duplicate records identified	1,376
Target column	status
Problem type	Binary classification

Target Variable

The target variable is status, which contains two classes:

- **Placed**: Candidate recorded as successfully placed.
- **Not Placed**: Candidate recorded as not placed.

Feature Description

The dataset contains different categories of candidate information.

Data Understanding and Quality Assessment

Before performing data cleaning, the dataset was examined to understand its structure, data types, missing values, duplicate records and inconsistencies.

This step is important because the quality of the input data directly affects machine learning model performance.

Missing Value Analysis

During the initial dataset inspection, missing values were identified in the following columns:

Column	Missing values
relocation_willingness	8,475
layoff_history	8,467
career_switch_willingness	8,466
relevant_experience	8,426
job_role_match	6,131

ssc_percentage	2,581
hsc_percentage	2,065
notice_period_days	1,582
employment_gap_months	1098

The missing values were handled using different approaches depending on the data type.

Duplicate Record Analysis

The dataset initially contained 1500 duplicate records.

Duplicate records were identified and removed to avoid repeated information influencing the analysis and model training.

Data Type and Consistency Checks

The dataset was also examined for:

- Incorrect data types.
- Inconsistent categorical values.
- Different uppercase and lowercase representations.
- Unnecessary spaces in categorical values.
- Invalid or unexpected values.

Categorical values were standardised to maintain consistency throughout the dataset.

Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed to prepare the dataset for exploratory analysis and machine learning. The following operations were carried out.

Removing Duplicate Records

Duplicate rows were identified and removed using duplicate detection techniques. This ensured that repeated records would not unnecessarily influence the model.

Handling Missing Values

Different imputation strategies were used for numerical and categorical columns.

Numerical columns

Missing values in the following columns were handled using median imputation:

- ssc_percentage
- hsc_percentage
- notice_period_days
- employment_gap_months

The median was selected because it is less affected by extreme values compared with the mean.

Categorical columns

Missing values in the following columns were handled using an Unknown category:

- layoff_history
- relocation_willingness
- career_switch_willingness
- relevant_experience
- job_role_match

Using an Unknown category helps retain records without assigning an incorrect category to missing information.

Standardising Categorical Values

Categorical values were checked and standardised to avoid duplicate categories caused by differences in text formatting.

For example:

- Yes
- YES
- yes

These values should be treated as the same category.

Exploratory Data Analysis (EDA)

EDA is used to understand the dataset, identify patterns, and explore relationships between candidate characteristics and placement outcomes.

Candidate Performance Analysis

- Compare academic scores between placed and non-placed candidates.
- Analyze technical, aptitude, and communication scores by placement status.
- Investigate whether stronger assessment scores are associated with placement

Skills and Interview Performance

- Examine the relationship between job-role matching and placement status.
- Compare candidate assessment scores across placement groups.
- Identify patterns that may help explain placement outcomes

Experience and Placement

- Analyze placement outcomes across experience levels.
- Compare candidates with and without internship experience.
- Examine whether work experience is associated with placement status.

Company Tier and Job-Related Factors

- Compare placement outcomes across company tiers.
- Examine expected compensation and placement status.
- Investigate the relationship between competition level and placement outcomes.

Correlation Analysis

A correlation heatmap can help identify relationships between numerical features.

Correlation indicates association, not necessarily causation. The analysis should not assume that one feature directly causes a candidate to be placed.

Feature Engineering

Feature engineering is the process of creating new features from existing columns to provide additional useful information to machine learning models.

In this project, five additional features were created.

Experience Category

A new feature named experience_category was created to group candidates based on their years of experience.

The categories include:

- Fresher
- Junior
- Senior

This feature helps represent candidate experience in a more understandable categorical format.

Academic Performance Band

A new feature named academic_performance_band was created using academic performance information.

Instead of relying only on individual percentage values, this feature groups candidates into broader academic performance categories.

Skills Match Level

The skills_match_level feature was created to represent how well the candidate's skills match the job requirements.

It provides a simplified representation of candidate suitability.

Average Interview Score

The average_interview_score feature was created by combining relevant assessment scores.

It provides an overall numerical representation of candidate assessment performance.

Interview Performance Category

The interview_performance_category feature was created using the average interview score.

It groups candidates into broader performance categories, making the assessment information easier for the model to interpret.

Machine Learning Model Development

After completing data cleaning, EDA and feature engineering, the prepared dataset was used to train supervised machine learning classification models.

Since the target variable contains two classes, the problem was treated as a binary classification problem.

Train-Test Split

The dataset was divided into training and testing sets. The training dataset is used to teach the model, while the testing dataset is used to evaluate how well the trained model performs on unseen records.

Data Preprocessing for Model Training

Numerical features – StandardScaler

StandardScaler was used to standardise numerical features where required. It transforms numerical values to have a mean of approximately zero and a standard deviation of approximately one.

This is particularly useful for algorithms such as KNN and SVM, which are sensitive to differences in feature scales.

Categorical features – OneHotEncoder

OneHotEncoder was used to convert categorical variables into numerical representations.

Machine Learning Algorithms Used

Four classification algorithms were explored and compared in this project.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. It estimates the probability that an observation belongs to a particular class.

In this project, Logistic Regression was used as a baseline classification model to predict whether a candidate would be Placed or Not Placed.

K-Nearest Neighbors (KNN)

KNN is a supervised machine learning algorithm that predicts the class of a new observation based on the classes of its nearest neighbours. It calculates distances between observations and identifies the closest data points. The majority class among the selected neighbours determines the prediction.

Decision Tree

A Decision Tree is a supervised machine learning algorithm that makes predictions using a sequence of decision rules. It divides the dataset into smaller groups based on feature values. The algorithm selects splits using criteria such as Gini Impurity or Entropy and Information Gain.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction performance. Instead of depending on one tree, it creates multiple trees using different samples and feature selections. Each tree makes a prediction, and the final classification is determined through majority voting.

Model Training and Selection

After exploring the five algorithms, their performance was evaluated using classification metrics. The objective was to identify a model that could provide reliable predictions for both Placed and Not Placed classes.

Model	Accuracy	Precision	Recall	F1_Score
Random Forest	89.41%	87.58%	75.74%	81.23%
Logistic Regression	88.14%	81.99%	77.92%	79.91%
Decision Tree	82.86%	71.01%	73.27%	72.12%
K-Nearest Neighbors	81.60%	68.69%	72.01%	70.31%

Random Forest was selected for further optimisation and final evaluation.

Streamlit Dashboard

The Streamlit dashboard provides an interactive interface for exploring candidate data and placement insights

Overview

The Overview section provides a summary of the dataset and placement information. It helps users understand the overall dataset at a glance.

Candidate Analysis

This section allows users to explore candidate information and understand different patterns in the dataset.

It can include:

- Academic scores versus placement status
- Placement Rate by Company Tier
- Certifications vs Placement
- Interview Performance vs Placement

Model Performance

This section presents the performance of the trained machine learning model.

It includes important evaluation metrics such as:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- ROC-AUC.
- Confusion matrix.
- Classification report.

This section helps users understand how effectively the model predicts placement outcomes.

Predict a Candidate

This is one of the most important sections of the application. It allows users to enter candidate information and obtain a placement prediction.

The predicted output is:

- Placed
- Not Placed

The application may also display the prediction probability to provide additional information about the model's confidence.

Important: Prediction probabilities are model estimates, not guarantees of actual placement outcomes.

Tools and Technologies

Technology	Purpose
Python	Main programming language

Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Matplotlib and Seaborn	Data visualization
Plotly	Interactive charts
Scikit-learn	Model training and evaluation
Streamlit	Interactive dashboard
Jupyter Notebook	Experimentation and documentation

Project Workflow

The complete project follows the workflow below:

Step 1: Data Collection

Load the HR job-placement dataset.

Step 2: Data Understanding

Understand the dataset structure, columns, data types and target variable.

Step 3: Data Cleaning

Remove duplicate records, handle missing values and standardise categorical values.

Step 4: Exploratory Data Analysis

Explore candidate attributes, target distribution and relationships between features.

Step 5: Feature Engineering

Create additional features to represent experience, academic performance, skill matching and interview performance.

Step 6: Data Preprocessing

Apply numerical scaling and categorical encoding as required.

Step 7: Train-Test Split

Split the dataset into 80% training and 20% testing data.

Step 8: Model Training

Train and compare five classification algorithms.

Step 9: Model Evaluation

Evaluate the models using accuracy, precision, recall, F1-score and ROC-AUC.

Step 10: Hyperparameter Tuning

Optimise the selected Random Forest model.

Step 11: Model Saving

Save the trained model for future use.

Step 12: Dashboard Development and Prediction

Build the interactive Streamlit application and Allow users to enter candidate information and generate placement predictions.

Conclusion

The Job Acceptance Prediction System was developed to analyse candidate information and predict their job placement status using machine learning.

In this project, we performed data cleaning, exploratory data analysis, feature engineering and model training. We explored five machine learning algorithms and selected the tuned Random Forest model for the final prediction. The selected model achieved approximately 89% accuracy and 0.96 ROC-AUC on the test dataset.

We also developed an interactive Streamlit dashboard that allows users to explore candidate data, view model performance and predict the placement status of new candidates.